

Activity 01

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-- Create Database
CREATE DATABASE SchoolDB_Advanced;
USE SchoolDB_Advanced;

-- 1. Teachers Table
CREATE TABLE Teachers (
    teacher_id INT PRIMARY KEY,
    teacher_name VARCHAR(50),
    email VARCHAR(100)
);

-- 2. Subjects Table
CREATE TABLE Subjects (
    subject_id INT PRIMARY KEY,
    subject_name VARCHAR(50),
    teacher_id INT,
    FOREIGN KEY (teacher_id) REFERENCES Teachers(teacher_id)
);

-- 3. Classes Table (New Table)
CREATE TABLE Classes (
    class_id INT PRIMARY KEY,
    class_name VARCHAR(20),
    class_teacher_id INT,
    FOREIGN KEY (class_teacher_id) REFERENCES Teachers(teacher_id)
);

-- 4. Students Table (Updated with class_id)
CREATE TABLE Students (
    student_id INT PRIMARY KEY,
    student_name VARCHAR(50),
    city VARCHAR(50),
    class_id INT,
    FOREIGN KEY (class_id) REFERENCES Classes(class_id)
);

-- 5. Exam_Marks Table
CREATE TABLE Exam_Marks (
    record_id INT PRIMARY KEY,
    student_id INT,
    subject_id INT,
    marks INT,
    FOREIGN KEY (student_id) REFERENCES Students(student_id),
    FOREIGN KEY (subject_id) REFERENCES Subjects(subject_id)
);

-- --- Inserting Sample Data ---

INSERT INTO Teachers VALUES
(1, 'Mr. Perera', 'perera@school.lk'),
(2, 'Mrs. Silva', 'silva@school.lk'),
(3, 'Mr. Gunawardena', 'guna@school.lk');
```

```
-- Create Database
CREATE DATABASE CompanyDB;
USE CompanyDB;
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-- 1. Locations Table
CREATE TABLE Locations (
    location_id INT PRIMARY KEY,
    city_name VARCHAR(50)
);

-- 2. Departments Table
CREATE TABLE Departments (
    dept_id INT PRIMARY KEY,
    dept_name VARCHAR(50),
    location_id INT,
    FOREIGN KEY (location_id) REFERENCES Locations(location_id)
);

-- 3. Job_Roles Table
CREATE TABLE Job_Roles (
    role_id INT PRIMARY KEY,
    role_title VARCHAR(50),
    basic_salary INT
);

-- 4. Employees Table
CREATE TABLE Employees (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    dept_id INT,
    role_id INT,
    FOREIGN KEY (dept_id) REFERENCES Departments(dept_id),
    FOREIGN KEY (role_id) REFERENCES Job_Roles(role_id)
);

-- 5. Projects Table
CREATE TABLE Projects (
    project_id INT PRIMARY KEY,
    project_name VARCHAR(50),
    dept_id INT,
    FOREIGN KEY (dept_id) REFERENCES Departments(dept_id)
);

-- 6. Project_Assignments Table
CREATE TABLE Project_Assignments (
    assignment_id INT PRIMARY KEY,
    emp_id INT,
    project_id INT,
    hours_worked INT,
    FOREIGN KEY (emp_id) REFERENCES Employees(emp_id),
    FOREIGN KEY (project_id) REFERENCES Projects(project_id)
);

-- --- Inserting Sample Data ---

INSERT INTO Locations VALUES
(1, 'Colombo'),
(2, 'Kandy'),
(3, 'Galle');
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INSERT INTO Departments VALUES
(10, 'IT Department', 1),
(20, 'HR Department', 2),
(30, 'Finance Department', 1);

INSERT INTO Job_Roles VALUES
(100, 'Software Engineer', 150000),
(101, 'HR Manager', 120000),
(102, 'Accountant', 100000),
(103, 'QA Engineer', 130000);

INSERT INTO Employees VALUES
(1, 'Nimal Perera', 10, 100), -- IT, SE
(2, 'Sunil Shantha', 10, 103), -- IT, QA
(3, 'Mala Silva', 20, 101), -- HR, Manager (No projects assigned)
(4, 'Kamal Fernando', 30, 102), -- Finance, Accountant
(5, 'Ruwan Kumara', 10, 100); -- IT, SE

INSERT INTO Projects VALUES
(1001, 'Website Redesign', 10),
(1002, 'Payroll System', 30),
(1003, 'Mobile App', 10),
(1004, 'Employee Portal', 20); -- No one assigned to this project yet

INSERT INTO Project_Assignments VALUES
(1, 1, 1001, 40), -- Nimal worked 40 hrs on Website
(2, 1, 1003, 20), -- Nimal worked 20 hrs on Mobile App
(3, 2, 1001, 35), -- Sunil worked 35 hrs on Website
(4, 4, 1002, 50), -- Kamal worked 50 hrs on Payroll
(5, 5, 1003, 45); -- Ruwan worked 45 hrs on Mobile App

```

Questions :

Q1: Show the names of all employees and their job titles.

Q2: Display the department names and the cities where they are located.

Q3: Show the employee names and the name of the department they work in.

Q4: Find the names of projects and the department names they belong to.

Q5: List the employee names and the hours they have worked on any project.

Q6: Show the names of employees, their job titles, and their department names.

Q7: Find the names of employees who work in the city of 'Colombo'.

Q8: List the project names and the names of employees assigned to them.

Q9: Show the names of 'Software Engineer's and the projects they are working on.

Q10: Calculate the total hours worked by each employee (Show Employee Name and Total Hours).

Q11: Show all employee names, their job titles, department names, and the city they work in.

Q12: Find the names of employees who have NOT been assigned to any project.

Q13: Show all projects and the total hours spent on them, even if no hours are logged yet.

Q14: Display the employee name, role title, project name, and hours worked.

Q15 (The Ultimate Query): The Grand Report - Show the Employee Name, Job Role, Department Name, City Name, Project Name, and Hours Worked in a single table! (Using ALL 6 TABLES)

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